



Intel StrataFlash® Embedded Memory (P30)

Delivering world class, price-performance flash solutions for embedded applications

Product Overview

Intel StrataFlash® Embedded Memory (P30) is Intel's lowest cost-per-bit NOR flash memory solution for the embedded market segment, manufactured on state-of-the-art 130 nm process lithography. P30 utilizes proven fourth-generation Intel StrataFlash® Memory—the world's leading multi-bit-per-cell technology. P30 meets the diverse requirements of embedded market segments such as consumer electronics, wired communications and industrial with a single chip code and data solution. Available in densities from 64 Mb to 1 Gb and in multiple packaging options, with high-performance access times, security modes and low power, the P30 is a powerful and compelling flash memory solution for embedded applications.



High Performance

P30 flash memory sets the standard for fast read speeds by offering synchronous burst mode and asynchronous page mode read operations that accelerate memory subsystems for exceptional performance. It offers design engineers multiple high-performance read modes to choose from, starting with 85 ns asynchronous access time to 40 MHz zero wait-state synchronous burst.

Multiple Densities and Packages

Densities available for the Intel StrataFlash Embedded Memory family range from 64 Mb up to 1 Gb, providing an easy upgrade path within each package family. Package offerings include TSOP, Easy BGA and Intel® QUAD+ SCSP. The 56-lead TSOP package is the industry standard for the embedded market segment, while the 64-ball Easy BGA package provides SOP reliability and long-term footprint compatibility in a chip-scale package size. The Intel QUAD+ SCSP packages offer small footprints for space-constrained applications.

Security Features

The P30 offers a number of features that provide customers with an additional level of security such as a robust individual zero-latency block locking scheme which protects against spurious writes. It also includes an optional, individual hardware lock using the WP# pin. For state-of-the-art security, P30 includes four blocks in the main array that can be made One-Time-Programmable (OTP). In addition, there are 2,112 user-programmable OTP register bits which can be used for security keys, traceability or other security needs.

Product Highlights

- Smallest 256 Mb NOR memory flash die in the industry
- Intel's lowest cost-per-bit product family for the embedded market segment
- Intel's proven fourth-generation Multi-Level Cell (MLC) technology, offering highly reliable operation
- Broad density offering: 64 Mb, 128 Mb, 256 Mb, 512 Mb and 1 Gb devices
- Extensive packaging options with TSOP, Easy BGA and Intel QUAD+ SCSP
- Available in Pb and Pb-free packaging
- Fast 85 ns asynchronous access time
- Burst read mode support at 40 MHz for fast code execution
- 1.8V VCC, 1.7-3.6V continuous I/O
- -40° C to +85° C operating temperatures
- Multiple security modes
- Minimum 100,000 cycles per block
- Royalty-free software support

Features	Benefits
Intel® Multi-Level Cell (MLC) technology	Exceptional value with proven fourth-generation technology
64 Mb, 128 Mb, 256 Mb, 512 Mb and 1 Gb densities	Lowest cost-per-bit NOR flash memory devices with full range of easy-to-upgrade, pin-compatible densities
56-Lead TSOP package	Industry standard for the embedded market segment
64-ball Easy BGA package	50% smaller than TSOP, greater security
Intel® QUAD+ Stacked-CSP package	Ideal for space-constrained applications
Burst and page mode interface	Fast code execution (40 MHz synchronous bus speed with zero wait-states)
Low-voltage 1.8V core	Ideal for battery-backed applications
Complete selection of royalty-free Intel® Flash management software	Dramatically reduces time-to-market for OEMs and is easily ported to OEM environments; manages code execution and data maintenance in one device
Common Flash Interface (CFI)	Compatibility with future products, today
One-Time Programmable (OTP) protection registers	64 bits programmed at the factory with unique ID; 2,112 bits can be programmed by OEM; traceability, license control, and system authentication
Selectable OTP	Four pre-defined 128 KB areas in the main array can be made OTP to support the highest levels of security
Buffered Enhanced Factory Programming (BEFP)	BEFP speeds up MLC flash programming for today's beat-rate sensitive manufacturing environments

Please go to www.intel.com/design/flash and click on "Documentation" from the left navigation for the most current version of the documents listed below:

Product Information

- Intel StrataFlash® Embedded Memory (P30) Product Brief
- Intel StrataFlash® Embedded Memory (P30) Datasheet

Software

- Intel® Flash Data Integrator Product Brief
- Intel® Persistent Storage Manager Product Brief
- Intel® Virtual Small Block File Manager Product Brief

Packaging

- Intel® Small Outline Package Guide
- Easy BGA Packaging for Intel® Flash Memory Devices
- Intel® Stacked Chip Scale Packaging Products

Related Information

- Intel® Flash Memory, The Smart Choice for Embedded Applications
- AP-793 Using Asynchronous Page and Synchronous Burst Read Modes with Intel® Flash Memory
- AP-812 Migration Guide for Intel StrataFlash® Memory (J3) to Intel StrataFlash® Embedded Memory (P30)
- AP-813 Migration Guide for Spansion® S29GLxxxN to Intel StrataFlash® Embedded Memory (P30)
- AP-825 Migration Guide for Intel StrataFlash® Synchronous Memory (K3/K18) to Intel StrataFlash® Embedded Memory (P30)

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